

Qualification Pack



Supervisor - Operations (Metro Rail)

Station Controller/ Train Operator

QP Code: DMR/LSC/Q0601

Version: 1.0

NSQF Level: 4.5

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DMR/LSC/Q0601: Supervisor - Operations (Metro Rail)

Brief Job Description

A Supervisor - Operations (Metro Rail) oversees metro railway operations, ensuring system efficiency, safety compliance, and smooth service delivery. The role includes managing signaling, telecom, and fare collection systems, supervising station operations, and handling rolling stock and train operations.

Personal Attributes

The individual should have strong analytical and problem-solving abilities to address operational challenges effectively. The role also requires excellent leadership and communication skills to guide teams and coordinate with stakeholders. Attention to detail and a commitment to safety and operational standards are essential for maintaining efficient metro railway operations.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [DMR/LSC/N9901: Implement metro operations training and safety procedures](#)
2. [DMR/LSC/N0601: Supervise the operations of signaling systems and platform screen doors](#)
3. [DMR/LSC/N0602: Supervise Telecom and Communication Systems in Metro Railways](#)
4. [DMR/LSC/N0603: Supervise metro train operations in Non-ATP modes](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Electives (mandatory to select at least one):

Elective 1: Station Controller

1. [DMR/LSC/N0604: Supervise Automated Fare Collection \(AFC\) system operations and maintenance](#)
2. [DMR/LSC/N0605: Supervise metro station operations](#)

Elective 2: Train Operator

1. [DMR/LSC/N0606: Supervise and troubleshoot rolling stock](#)
2. [DMR/LSC/N0607: Supervise SOP train operations and practical tasks](#)
3. [DMR/LSC/N0608: Operate metro train on test track, mainline \(non-revenue hours\), and simulator](#)

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Qualification Pack (QP) Parameters

Sector	Logistics
Sub-Sector	Transport (Metro Rail)
Occupation	Metro Rail Operations
Country	India
NSQF Level	4.5
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/1324.1300
Minimum Educational Qualification & Experience	Diploma (3 years Engineering Diploma* / equivalent OR Graduate** *Diploma in Electrical/Electronic or equivalent **B Sc. Hons in (Physics/ Chemistry/ Mathematics) OR B Sc. (Physics/ Chemistry/ Mathematics) from a Govt. recognized University/Institute)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	08/05/2028
NSQC Approval Date	08/05/2025
Version	1.0
Reference code on NQR	QG-4.5-TW-04145-2025-V1-DMRCA
NQR Version	1.0

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DMR/LSC/N9901: Implement metro operations training and safety procedures

Description

This unit focuses on implementing structured training and safety measures for metro operations, ensuring technical knowledge, system readiness, and emergency preparedness.

Scope

The scope covers the following :

- Training Module Overview
- Safety Overview
- Store Management and Procurement
- Overview of Telecom, Signaling, and Track Systems
- Overview of Traction, SCADA, and E&M Systems
- OCC and Rolling Stock Overview
- Operations and Customer Care
- Practical Visits
- First Aid Training
- Fire Fighting Systems
- Security in Metro Systems
- Metro Rail General Rules 2020
- Safety Circulars and Contingency Plans

Elements and Performance Criteria

Training Module Overview

To be competent, the user/individual on the job must be able to:

- PC1.** follow schedules for weekly activities and guidelines on punctuality, discipline, and conduct.
- PC2.** conduct identity verification and maintain attendance records
- PC3.** ensure adherence to training timelines and program deliverables

Safety Overview

To be competent, the user/individual on the job must be able to:

- PC4.** identify the role of the safety department and follow measures for personal, passenger, and system safety
- PC5.** use appropriate Personal Protective Equipment (PPE) in all designated areas
- PC6.** follow the hazard identification and risk assessment processes

Store Management and Procurement

To be competent, the user/individual on the job must be able to:

- PC7.** perform store management activities, including stock handling, documentation, and interdepartmental coordination for procurement.
- PC8.** monitor inventory levels and ensure timely requisitions for required materials
- PC9.** maintain records of procurement processes and vendor communication

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Overview of Telecom, Signaling, and Track Systems

To be competent, the user/individual on the job must be able to:

- PC10.** identify telecom components (AFC, CCTV, PIDS, radio systems, clocks) and perform basic functional checks
- PC11.** identify signaling systems (CATC, CBTC) and observe/report signal functioning during visits.
- PC12.** identify P-Way components (rails, sleepers, crossings) and report track conditions through basic patrolling tasks
- PC13.** report malfunctions in telecom, signaling, or track systems using the correct reporting protocols

Overview of Traction, SCADA, and E&M Systems

To be competent, the user/individual on the job must be able to:

- PC14.** identify traction power and SCADA components and interpret sectioning diagrams
- PC15.** identify E&M systems (RSS, ASS, DG sets, UPS) and perform basic checks to understand their functionality
- PC16.** report unusual system performance and assist in routine inspections

OCC and Rolling Stock Overview

To be competent, the user/individual on the job must be able to:

- PC17.** familiarize with OCC systems, tools, and the role of OCC controllers
- PC18.** identify rolling stock components and their maintenance procedures
- PC19.** report rolling stock irregularities observed during operations

Operations and Customer Care

To be competent, the user/individual on the job must be able to:

- PC20.** familiarize with station operations, revenue management, and crew control functions
- PC21.** identify public complaint channels, possible causes, and resolve complaints using proper processes
- PC22.** assist in managing passenger inquiries and ensuring customer satisfaction

Practical Visits

To be competent, the user/individual on the job must be able to:

- PC23.** conduct visits to OCC, stations, depots, and tunnels to observe layouts, facilities, and operations
- PC24.** document key observations during practical visits for knowledge reinforcement
- PC25.** identify operational challenges during visits and suggest improvement areas

First Aid Training

To be competent, the user/individual on the job must be able to:

- PC26.** identify the role and scope of first aid, including key human body functions (e.g., circulatory, respiratory systems)
- PC27.** follow appropriate techniques for treating bleeding, fractures, burns, shock, and unconsciousness
- PC28.** perform artificial respiration, CPR, and safe transportation of injured persons using stretchers
- PC29.** participate in simulated first aid scenarios to enhance preparedness

Fire Fighting Systems

To be competent, the user/individual on the job must be able to:

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- PC30.** identify elements of combustion, classes of fire, and appropriate extinguishing methods
- PC31.** operate active/passive fire protection systems, including alarms, fire extinguishers, and fire hydrants
- PC32.** conduct practical drills and identify fire suppression equipment at metro stations
- PC33.** respond to fire incidents promptly, ensuring passenger and system safety

Security in Metro Systems

To be competent, the user/individual on the job must be able to:

- PC34.** identify security threats, red alert situations, and follow protocols for disaster management
- PC35.** handle theft cases, manage abandoned persons, and lodge FIRs as per protocol
- PC36.** coordinate with security personnel to mitigate security breaches effectively

Metro Rail General Rules 2020

To be competent, the user/individual on the job must be able to:

- PC37.** interpret key provisions under Metro Rail General Rules 2020, including signals, train speed limits, and safety protocols
- PC38.** follow operational procedures for maintenance, station control duties, and driverless train systems
- PC39.** ensure compliance with rules for safe and efficient train operations

Safety Circulars and Contingency Plans

To be competent, the user/individual on the job must be able to:

- PC40.** implement procedures for the imposition of TSR, axle counter reset, Train ID management, and Computer Aided Train Supervision (CATS) operation
- PC41.** follow guidelines for engineering possession and maintenance activities during non-revenue hours, including hot work, solar system testing/repair, and role-play exercises
- PC42.** ensure compliance with procedures for stopping trains at platforms and during operational disruptions, such as earthquakes, high winds, and floods
- PC43.** manage degraded mode operations, including train movement during interlocking failures
- PC44.** oversee the issue, revision, updating, and deletion of SOPs, SWO/DWO, and timetable trials for new sections
- PC45.** facilitate the initiation of revenue train services, including pilot train movement, ATO/UTO/ATB mode operations, and AFTC functionality checks
- PC46.** execute procedures for negotiating neutral sections, managing pantograph failures, and monitoring Panto-OHE interface cameras
- PC47.** oversee train stabling, depot entry/exit, train movement, and fitness checks
- PC48.** conduct emergency evacuation using UG facilities, mid-tunnel shafts, and walkways for detainment of passengers due to technical failure, explosion, or fire
- PC49.** operate and manage non-ATP vehicles such as RGM and CMV/CTMC and address incidents like rail fractures, weld failures, rollback scenarios, radio communication issues and resetting of passenger alarms in train
- PC50.** coordinate vehicle movement on link lines and rescue immobilized trains
- PC51.** respond to fire/smoke incidents, including UG train fires, ensuring passenger and staff safety
- PC52.** operate ETS/BLS plungers, remove foreign objects like kites from OHE, and follow OHE safety guidelines

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of punctuality, discipline, and structured training
- KU2.** the procedures for trainee verification and attendance record maintenance
- KU3.** the role of time management in achieving training milestones
- KU4.** the role of the safety department and station/depot safety protocols
- KU5.** the use and importance of PPE
- KU6.** the basics of hazard identification, risk management, and accident reporting
- KU7.** the basics of stores management (inventory control, documentation)
- KU8.** the procurement processes and coordination across departments
- KU9.** the procedures for vendor management and maintaining procurement transparency
- KU10.** the telecom components, including AFC, CCTV, clocks, and communication systems
- KU11.** the basics of signaling systems (CATC, CBTC) and reporting unusual signal observations
- KU12.** the key P-Way components, track patrolling, and reporting track conditions
- KU13.** the common malfunctions in telecom, signaling, and track systems
- KU14.** the traction power systems and sectioning diagrams
- KU15.** the importance of SCADA and its functionalities
- KU16.** the role of E&M systems (DG sets, UPS) and their use during emergencies
- KU17.** the key preventive maintenance checks for E&M systems
- KU18.** the OCC systems, tools, and controller responsibilities
- KU19.** the rolling stock components, features, and basic maintenance
- KU20.** the reporting and addressing of common rolling stock issues
- KU21.** the station operations, revenue management, and crew coordination
- KU22.** the causes of public complaints and grievance resolution processes
- KU23.** the techniques for effective customer service and passenger engagement
- KU24.** the observations and learning outcomes from OCC, station, depot, and tunnel visits
- KU25.** the identifying real-world challenges during visits and linking theory to practice
- KU26.** the basic principles of first aid and key human body functions
- KU27.** the management of bleeding, fractures, burns, and unconsciousness
- KU28.** the artificial respiration, CPR, and safe patient transportation
- KU29.** the importance of first aid preparedness during emergencies
- KU30.** the elements of combustion, fire classifications, and extinguishing methods
- KU31.** the principles of passive/active fire protection systems
- KU32.** the practical usage of fire extinguishers, hydrants, and drills
- KU33.** the emergency response procedures during fire incidents
- KU34.** the security protocols, red alert situations, and theft management processes
- KU35.** the protocols for managing abandoned persons and disaster situations
- KU36.** the role of security personnel in ensuring passenger and asset safety
- KU37.** the key terminologies, signal usage, and operational protocols

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- KU38.** the rules for maintenance works, train operations, and driverless train systems
- KU39.** the role of regulatory compliance in metro rail safety
- KU40.** the safety circulars and contingency plans for fire/security incidents
- KU41.** the importance of proactive contingency planning for emergency scenarios.
- KU42.** the importance role-playing and drills to improve safety and response performance
- KU43.** the procedures for TSR imposition, axle counter reset, Train ID management, and CATS operations
- KU44.** the engineering possession protocols for maintenance tasks during non-revenue hours, including solar system testing and hot work
- KU45.** the guidelines for safely halting train operations during emergencies, natural disasters, or technical failures
- KU46.** the steps for managing degraded mode operations and interlocking failures
- KU47.** the SOP and SWO/DWO lifecycle management, including revisions and timetable trials for new sections
- KU48.** the ATO, UTO, and ATB mode operations, emphasizing operational readiness and AFTC checks
- KU49.** the procedures for neutral section negotiation, pantograph damage management, and monitoring Panto-OHE interfaces using cameras
- KU50.** the best practices for train stabling, depot operations, and train fitness verification
- KU51.** the emergency evacuation protocols, focusing on passenger safety in UG and mid-tunnel situations
- KU52.** the guidelines for operating non-ATP vehicles, handling rail fractures, weld failures, and resetting alarms during rollbacks
- KU53.** the vehicle operations on link lines and immobilized train rescue procedures
- KU54.** the protocols for fire/smoke incidents, particularly UG fires, and ensuring safety during such incidents
- KU55.** the operation of ETS/BLS plungers and removal of foreign objects from OHE systems, along with OHE safety practices

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** convey clear and concise information to teams, junior engineers, and customers
- GS2.** effectively prioritize tasks and adhere to training schedules and operational timelines
- GS3.** identify operational challenges and propose practical solutions during training or real-time operations
- GS4.** assess system performance, identify malfunctions, and report anomalies accurately
- GS5.** work cohesively with peers, instructors, and junior engineers to meet training objectives
- GS6.** address passenger queries and manage grievances professionally
- GS7.** proactively identify risks, follow safety protocols, and contribute to hazard-free operations
- GS8.** adapt to diverse training environments, system layouts, and evolving operational scenarios
- GS9.** accurately maintain logs, records, and reports for training and practical activities

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- GS10.** monitor systems, equipment, and processes, identifying areas requiring intervention or improvement

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Training Module Overview</i>	2	2	-	1
PC1. follow schedules for weekly activities and guidelines on punctuality, discipline, and conduct.	-	-	-	-
PC2. conduct identity verification and maintain attendance records	-	-	-	-
PC3. ensure adherence to training timelines and program deliverables	-	-	-	-
<i>Safety Overview</i>	2	2	-	1
PC4. identify the role of the safety department and follow measures for personal, passenger, and system safety	-	-	-	-
PC5. use appropriate Personal Protective Equipment (PPE) in all designated areas	-	-	-	-
PC6. follow the hazard identification and risk assessment processes	-	-	-	-
<i>Store Management and Procurement</i>	2	2	-	1
PC7. perform store management activities, including stock handling, documentation, and interdepartmental coordination for procurement.	-	-	-	-
PC8. monitor inventory levels and ensure timely requisitions for required materials	-	-	-	-
PC9. maintain records of procurement processes and vendor communication	-	-	-	-
<i>Overview of Telecom, Signaling, and Track Systems</i>	2	2	-	1
PC10. identify telecom components (AFC, CCTV, PIDS, radio systems, clocks) and perform basic functional checks	-	-	-	-
PC11. identify signaling systems (CATC, CBTC) and observe/report signal functioning during visits.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. identify P-Way components (rails, sleepers, crossings) and report track conditions through basic patrolling tasks	-	-	-	-
PC13. report malfunctions in telecom, signaling, or track systems using the correct reporting protocols	-	-	-	-
<i>Overview of Traction, SCADA, and E&M Systems</i>	2	2	-	1
PC14. identify traction power and SCADA components and interpret sectioning diagrams	-	-	-	-
PC15. identify E&M systems (RSS, ASS, DG sets, UPS) and perform basic checks to understand their functionality	-	-	-	-
PC16. report unusual system performance and assist in routine inspections	-	-	-	-
<i>OCC and Rolling Stock Overview</i>	2	2	-	1
PC17. familiarize with OCC systems, tools, and the role of OCC controllers	-	-	-	-
PC18. identify rolling stock components and their maintenance procedures	-	-	-	-
PC19. report rolling stock irregularities observed during operations	-	-	-	-
<i>Operations and Customer Care</i>	2	2	-	1
PC20. familiarize with station operations, revenue management, and crew control functions	-	-	-	-
PC21. identify public complaint channels, possible causes, and resolve complaints using proper processes	-	-	-	-
PC22. assist in managing passenger inquiries and ensuring customer satisfaction	-	-	-	-
<i>Practical Visits</i>	1	1	-	0.5
PC23. conduct visits to OCC, stations, depots, and tunnels to observe layouts, facilities, and operations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. document key observations during practical visits for knowledge reinforcement	-	-	-	-
PC25. identify operational challenges during visits and suggest improvement areas	-	-	-	-
<i>First Aid Training</i>	1	1	-	0.5
PC26. identify the role and scope of first aid, including key human body functions (e.g., circulatory, respiratory systems)	-	-	-	-
PC27. follow appropriate techniques for treating bleeding, fractures, burns, shock, and unconsciousness	-	-	-	-
PC28. perform artificial respiration, CPR, and safe transportation of injured persons using stretchers	-	-	-	-
PC29. participate in simulated first aid scenarios to enhance preparedness	-	-	-	-
<i>Fire Fighting Systems</i>	1	1	-	0.5
PC30. identify elements of combustion, classes of fire, and appropriate extinguishing methods	-	-	-	-
PC31. operate active/passive fire protection systems, including alarms, fire extinguishers, and fire hydrants	-	-	-	-
PC32. conduct practical drills and identify fire suppression equipment at metro stations	-	-	-	-
PC33. respond to fire incidents promptly, ensuring passenger and system safety	-	-	-	-
<i>Security in Metro Systems</i>	1	1	-	0.5
PC34. identify security threats, red alert situations, and follow protocols for disaster management	-	-	-	-
PC35. handle theft cases, manage abandoned persons, and lodge FIRs as per protocol	-	-	-	-
PC36. coordinate with security personnel to mitigate security breaches effectively	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Metro Rail General Rules 2020</i>	1	1	-	0.5
PC37. interpret key provisions under Metro Rail General Rules 2020, including signals, train speed limits, and safety protocols	-	-	-	-
PC38. follow operational procedures for maintenance, station control duties, and driverless train systems	-	-	-	-
PC39. ensure compliance with rules for safe and efficient train operations	-	-	-	-
<i>Safety Circulars and Contingency Plans</i>	1	1	-	0.5
PC40. implement procedures for the imposition of TSR, axle counter reset, Train ID management, and Computer Aided Train Supervision (CATS) operation	-	-	-	-
PC41. follow guidelines for engineering possession and maintenance activities during non-revenue hours, including hot work, solar system testing/repair, and role-play exercises	-	-	-	-
PC42. ensure compliance with procedures for stopping trains at platforms and during operational disruptions, such as earthquakes, high winds, and floods	-	-	-	-
PC43. manage degraded mode operations, including train movement during interlocking failures	-	-	-	-
PC44. oversee the issue, revision, updating, and deletion of SOPs, SWO/DWO, and timetable trials for new sections	-	-	-	-
PC45. facilitate the initiation of revenue train services, including pilot train movement, ATO/UTO/ATB mode operations, and AFTC functionality checks	-	-	-	-
PC46. execute procedures for negotiating neutral sections, managing pantograph failures, and monitoring Panto-OHE interface cameras	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC47. oversee train stabling, depot entry/exit, train movement, and fitness checks	-	-	-	-
PC48. conduct emergency evacuation using UG facilities, mid-tunnel shafts, and walkways for detrainment of passengers due to technical failure, explosion, or fire	-	-	-	-
PC49. operate and manage non-ATP vehicles such as RGM and CMV/CTMC and address incidents like rail fractures, weld failures, rollback scenarios, radio communication issues and resetting of passenger alarms in train	-	-	-	-
PC50. coordinate vehicle movement on link lines and rescue immobilized trains	-	-	-	-
PC51. respond to fire/smoke incidents, including UG train fires, ensuring passenger and staff safety	-	-	-	-
PC52. operate ETS/BLS plungers, remove foreign objects like kites from OHE, and follow OHE safety guidelines	-	-	-	-
NOS Total	20	20	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N9901
NOS Name	Implement metro operations training and safety procedures
Sector	Logistics
Sub-Sector	
Occupation	Foundation – Metro (Rail)
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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DMR/LSC/N0601: Supervise the operations of signaling systems and platform screen doors

Description

This unit covers supervisory tasks related to the operation, maintenance, and troubleshooting of metro railway signaling systems, including Automatic Train Supervision (ATS), Automatic Train Protection (ATP), and Platform Screen Doors (PSD), ensuring seamless and safe railway operations.

Scope

The scope covers the following :

- Basic Signaling
- Signaling ATS
- Signaling ATP
- CCTV (ATS)
- Platform Screen Doors (PSD)

Elements and Performance Criteria

Basic Signaling

To be competent, the user/individual on the job must be able to:

- PC1.** identify signaling equipment, marker boards, and their functions on-site
- PC2.** verify interlocking systems to ensure safe train movements
- PC3.** perform system checks for primary and secondary train detection systems
- PC4.** perform fault detection and report issues in train detection systems

Signaling ATS

To be competent, the user/individual on the job must be able to:

- PC5.** operate ATS systems to execute commands and monitor indications
- PC6.** respond to abnormalities in ATS indications
- PC7.** execute Visual Display Unit (VDU) commands and monitor their outcomes
- PC8.** verify system responses to executed VDU commands and resolve inconsistencies
- PC9.** execute practical demonstrations of signaling commands using ATS/VDU systems
- PC10.** monitor and document outcomes of signaling commands for operational accuracy

Signaling ATP

To be competent, the user/individual on the job must be able to:

- PC11.** familiarize with the operation of ATP hardware components and Driver Machine Interface (DMI)
- PC12.** execute driving mode selection and apply Automatic Train Control (ATC) safety features
- PC13.** carry out operational procedures in degraded mode while ensuring safety standards
- PC14.** supervise train operations in mainline and depot areas, ensuring adherence to safety protocols
- PC15.** perform troubleshooting to resolve common ATP faults and restore functionality

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PC16. carry out assessments with trackside ATP equipment

PC17. perform operational checks on train-borne ATP equipment

PC18. execute operational tests on DMI indications and provide guidance to drivers

CCTV (ATS)

To be competent, the user/individual on the job must be able to:

PC19. operate CCTV systems to monitor train locations and movements

PC20. respond to emergencies using CCTV systems to mitigate risks and ensure safety

Platform Screen Doors (PSD)

To be competent, the user/individual on the job must be able to:

PC21. familiarize with PSD layout, numbering, and numbering for accurate supervision

PC22. verify the PSD layout and execute manual operations of PSD in case of system failure

PC23. use safety labels and special keys to operate PSD as per standard procedures

PC24. supervise adherence to PSD operating staff procedures during normal and emergency operations

PC25. carry out troubleshooting to identify and resolve PSD malfunctions effectively

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the basics of metro signaling concepts, CBTC, and S&T systems

KU2. the operational principles of metro railway signaling systems

KU3. the functions and handling of signaling equipment, marker boards, and interlocking systems

KU4. the train detection systems and fault diagnostics

KU5. the ATS architecture, command execution, and troubleshooting

KU6. the ATP hardware, DMI operations, and safety protocols

KU7. the functions of degraded modes and their handling procedures

KU8. the CCTV usage for train monitoring and emergency responses

KU9. the PSD layout, operation, and troubleshooting methods

KU10. the safety standards and compliance protocols for metro railway operations

KU11. the guidelines for emergency management and degraded mode operations

KU12. the communication protocols for coordination during routine and emergency operations

KU13. the environmental and occupational safety guidelines for railway operations

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. communicate effectively with team members and stakeholders during operational tasks

GS2. resolve operational and technical issues in signaling systems and PSDs

GS3. make informed decisions to address abnormalities and emergencies

GS4. analyze system feedback and implement corrective actions efficiently

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- GS5.** supervise and guide technical staff for seamless operations
- GS6.** manage time effectively to prioritize and execute tasks under tight schedules
- GS7.** apply technical knowledge in practical scenarios for fault detection and resolution
- GS8.** adjust to changing conditions and degraded modes in operations
- GS9.** monitor system operations to detect and address anomalies
- GS10.** train junior staff on the operation and troubleshooting of signaling systems and PSDs

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Basic Signaling</i>	4	4	-	2
PC1. identify signaling equipment, marker boards, and their functions on-site	-	-	-	-
PC2. verify interlocking systems to ensure safe train movements	-	-	-	-
PC3. perform system checks for primary and secondary train detection systems	-	-	-	-
PC4. perform fault detection and report issues in train detection systems	-	-	-	-
<i>Signaling ATS</i>	4	4	-	2
PC5. operate ATS systems to execute commands and monitor indications	-	-	-	-
PC6. respond to abnormalities in ATS indications	-	-	-	-
PC7. execute Visual Display Unit (VDU) commands and monitor their outcomes	-	-	-	-
PC8. verify system responses to executed VDU commands and resolve inconsistencies	-	-	-	-
PC9. execute practical demonstrations of signaling commands using ATS/VDU systems	-	-	-	-
PC10. monitor and document outcomes of signaling commands for operational accuracy	-	-	-	-
<i>Signaling ATP</i>	4	4	-	2
PC11. familiarize with the operation of ATP hardware components and Driver Machine Interface (DMI)	-	-	-	-
PC12. execute driving mode selection and apply Automatic Train Control (ATC) safety features	-	-	-	-
PC13. carry out operational procedures in degraded mode while ensuring safety standards	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. supervise train operations in mainline and depot areas, ensuring adherence to safety protocols	-	-	-	-
PC15. perform troubleshooting to resolve common ATP faults and restore functionality	-	-	-	-
PC16. carry out assessments with trackside ATP equipment	-	-	-	-
PC17. perform operational checks on train-borne ATP equipment	-	-	-	-
PC18. execute operational tests on DMI indications and provide guidance to drivers	-	-	-	-
CCTV (ATS)	4	4	-	2
PC19. operate CCTV systems to monitor train locations and movements	-	-	-	-
PC20. respond to emergencies using CCTV systems to mitigate risks and ensure safety	-	-	-	-
Platform Screen Doors (PSD)	4	4	-	2
PC21. familiarize with PSD layout, numbering, and numbering for accurate supervision	-	-	-	-
PC22. verify the PSD layout and execute manual operations of PSD in case of system failure	-	-	-	-
PC23. use safety labels and special keys to operate PSD as per standard procedures	-	-	-	-
PC24. supervise adherence to PSD operating staff procedures during normal and emergency operations	-	-	-	-
PC25. carry out troubleshooting to identify and resolve PSD malfunctions effectively	-	-	-	-
NOS Total	20	20	-	10

Logo
Not
Available



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0601
NOS Name	Supervise the operations of signaling systems and platform screen doors
Sector	Logistics
Sub-Sector	
Occupation	Metro Rail Operations
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DMR/LSC/N0602: Supervise Telecom and Communication Systems in Metro Railways

Description

This unit is about operating and managing communication systems essential for metro railway.

Scope

The scope covers the following :

- Train Radio Operations
- Hand portable Devices & RAU (Zetron)
- Emergency Communication Procedures
- CCTV Operations
- Communication Practices Using Hand portable Train Radio
- PIDS/PAS Systems

Elements and Performance Criteria

Train Radio Operations

To be competent, the user/individual on the job must be able to:

- PC1.** identify and operate various modules and keys on the train radio system
- PC2.** initiate and manage various call types and group communication modes using the train radio
- PC3.** use key-buttons on TRCP, RCH, and handsets appropriately for seamless communication

Handportable Devices & RAU (Zetron)

To be competent, the user/individual on the job must be able to:

- PC4.** use handportable devices and RAU (Zetron) properly for communication tasks
- PC5.** operate different keys and functions effectively on handportable devices and RAU (Zetron)

Emergency Communication Procedures

To be competent, the user/individual on the job must be able to:

- PC6.** utilize appropriate communication protocols to convey emergency information clearly and promptly
- PC7.** identify and operate available emergency communication systems to manage and resolve incidents

CCTV Operations

To be competent, the user/individual on the job must be able to:

- PC8.** monitor and operate CCTV systems to ensure continuous surveillance and security within the metro railway

Communication Practices Using Handportable Train Radio

To be competent, the user/individual on the job must be able to:

- PC9.** communicate procedures and actions effectively during door failure incidents using handportable/train radio

Qualification Pack

PC10. utilize communication systems to respond to and manage passenger alarm activations appropriately

PC11. maintain effective communication during degraded mode operations and other system failures using handportable/train radio

PIDS/PAS Systems

To be competent, the user/individual on the job must be able to:

PC12. operate Passenger Information Display System (PIDS) to display accurate and timely passenger information

PC13. utilize Passenger Announcement System (PAS) to make clear and audible announcements to passengers

PC14. navigate and manage system interfaces to ensure seamless operation of PIDS/PAS

PC15. switch between and manage different operational modes of PIDS/PAS systems as required

PC16. display various types of messages accurately and perform announcements using PIDS/PAS systems

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the types and functionalities of train radios, hand portable devices, RAU (Zetron), CCTV, PIDS/PAS systems

KU2. the modules, keys, and functions of train radio equipment including TRCP, RCH, and handsets

KU3. the procedures and systems for emergency communication within metro operations

KU4. the operational aspects of CCTV systems in monitoring and security

KU5. the standard operating procedures for effective communication during normal operations and emergencies

KU6. the functionality and interfaces of PIDS and PAS

KU7. the safety protocols and standards related to the operation of communication systems in metro environments

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. exchange information effectively using various communication systems

GS2. identify and resolve issues related to communication equipment and systems

GS3. collaborate with team members during communication operations and emergency responses

GS4. manage communication tasks efficiently within operational schedules

GS5. adjust communication practices in varying scenarios and under pressure

GS6. assess situations to determine appropriate communication actions and solutions

GS7. prioritize and adhere to safety protocols in all communication operations

GS8. provide clear and accurate passenger information through communication systems

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Train Radio Operations</i>	3	3	-	2
PC1. identify and operate various modules and keys on the train radio system	-	-	-	-
PC2. initiate and manage various call types and group communication modes using the train radio	-	-	-	-
PC3. use key-buttons on TRCP, RCH, and handsets appropriately for seamless communication	-	-	-	-
<i>Handportable Devices & RAU (Zetron)</i>	3	3	-	2
PC4. use handportable devices and RAU (Zetron) properly for communication tasks	-	-	-	-
PC5. operate different keys and functions effectively on handportable devices and RAU (Zetron)	-	-	-	-
<i>Emergency Communication Procedures</i>	3	3	-	1
PC6. utilize appropriate communication protocols to convey emergency information clearly and promptly	-	-	-	-
PC7. identify and operate available emergency communication systems to manage and resolve incidents	-	-	-	-
<i>CCTV Operations</i>	3	3	-	1
PC8. monitor and operate CCTV systems to ensure continuous surveillance and security within the metro railway	-	-	-	-
<i>Communication Practices Using Handportable Train Radio</i>	4	4	-	2
PC9. communicate procedures and actions effectively during door failure incidents using handportable/train radio	-	-	-	-
PC10. utilize communication systems to respond to and manage passenger alarm activations appropriately	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. maintain effective communication during degraded mode operations and other system failures using handportable/train radio	-	-	-	-
<i>PIDS/PAS Systems</i>	4	4	-	2
PC12. operate Passenger Information Display System (PIDS) to display accurate and timely passenger information	-	-	-	-
PC13. utilize Passenger Announcement System (PAS) to make clear and audible announcements to passengers	-	-	-	-
PC14. navigate and manage system interfaces to ensure seamless operation of PIDS/PAS	-	-	-	-
PC15. switch between and manage different operational modes of PIDS/PAS systems as required	-	-	-	-
PC16. display various types of messages accurately and perform announcements using PIDS/PAS systems	-	-	-	-
NOS Total	20	20	-	10

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Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0602
NOS Name	Supervise Telecom and Communication Systems in Metro Railways
Sector	Logistics
Sub-Sector	
Occupation	Metro Rail Operations
NSQF Level	4.5
Credits	0.5
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DMR/LSC/N0603: Supervise metro train operations in Non-ATP modes

Description

This unit outlines the competencies required to operate metro trains efficiently and safely in non-ATP (Automatic Train Protection) modes, focusing on practical tasks, decision-making, and risk management.

Scope

The scope covers the following :

- Train Operation in Non-ATP Modes
- ATP-related Procedures and Role Play
- Case Study on Non-ATP Mode and Role Play
- Simulator for Train Operation in Non-ATP Modes

Elements and Performance Criteria

Train Operation in Non-ATP Modes

To be competent, the user/individual on the job must be able to:

- PC1.** inspect and prepare train systems for non-ATP operation, ensuring manual overrides and safety systems are functional
- PC2.** transition to non-ATP mode following standard procedures and communicate with the control center for guidance
- PC3.** operate trains manually, maintaining prescribed speed limits, interpreting signals, and responding to trackside indicators
- PC4.** ensure safe braking, acceleration, and alignment with platforms to facilitate passenger boarding and alighting
- PC5.** manage emergencies during non-ATP operations, such as system failures, signal loss, or unexpected obstacles

ATP-related Procedure and Role Play

To be competent, the user/individual on the job must be able to:

- PC6.** demonstrate emergency response actions, including train stoppage, passenger communication, and coordination with control
- PC7.** participate in role-play scenarios to simulate non-ATP operation challenges, focusing on communication and decision-making

Case Study on Non-ATP Mode and Role Play

To be competent, the user/individual on the job must be able to:

- PC8.** analyze case studies to identify risks, best practices, and lessons learned in non-ATP scenarios
- PC9.** apply knowledge from case studies to role-play exercises, practicing situational awareness and quick decision-making

Simulator for Train Operation in Non-ATP Modes

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC10.** operate a train simulator to practice manual operations, including braking, acceleration, and responding to simulated emergencies
- PC11.** use simulator feedback to improve operational skills, focusing on safety and efficiency

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the types and functionalities of train control systems, including ATP and non-ATP modes
- KU2.** the standard operating procedures for Restricted Manual, Coded Manual, and Non-Coded Manual modes
- KU3.** the emergency response protocols and troubleshooting methods for non-ATP conditions
- KU4.** the safety regulations and guidelines for metro railway operations
- KU5.** the communication systems and protocols with OCC and field staff
- KU6.** the case histories of non-ATP incidents and lessons learned
- KU7.** the features and usage of train operation simulators
- KU8.** the passenger safety and crowd management techniques
- KU9.** the operational limits such as speed, braking distance, and signal interpretations
- KU10.** the documentation and reporting requirements for non-ATP operations

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** make prompt and effective decisions during non-ATP scenarios
- GS2.** troubleshoot system failures and operational issues in real-time
- GS3.** convey critical information clearly to OCC, field staff, and passengers
- GS4.** work effectively with the OCC and other teams during degraded operations
- GS5.** manage schedules and delays efficiently in non-ATP modes
- GS6.** adjust to dynamic situations and varying operating conditions
- GS7.** monitor systems and signals accurately to ensure operational safety
- GS8.** evaluate case studies and simulation results for continuous improvement
- GS9.** guide the onboard team and passengers during emergencies
- GS10.** operate and troubleshoot simulators and train systems competently

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Train Operation in Non-ATP Modes</i>	5	5	-	4
PC1. inspect and prepare train systems for non-ATP operation, ensuring manual overrides and safety systems are functional	-	-	-	-
PC2. transition to non-ATP mode following standard procedures and communicate with the control center for guidance	-	-	-	-
PC3. operate trains manually, maintaining prescribed speed limits, interpreting signals, and responding to trackside indicators	-	-	-	-
PC4. ensure safe braking, acceleration, and alignment with platforms to facilitate passenger boarding and alighting	-	-	-	-
PC5. manage emergencies during non-ATP operations, such as system failures, signal loss, or unexpected obstacles	-	-	-	-
<i>ATP-related Procedure and Role Play</i>	5	5	-	2
PC6. demonstrate emergency response actions, including train stoppage, passenger communication, and coordination with control	-	-	-	-
PC7. participate in role-play scenarios to simulate non-ATP operation challenges, focusing on communication and decision-making	-	-	-	-
<i>Case Study on Non-ATP Mode and Role Play</i>	5	5	-	2
PC8. analyze case studies to identify risks, best practices, and lessons learned in non-ATP scenarios	-	-	-	-
PC9. apply knowledge from case studies to role-play exercises, practicing situational awareness and quick decision-making	-	-	-	-
<i>Simulator for Train Operation in Non-ATP Modes</i>	5	5	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. operate a train simulator to practice manual operations, including braking, acceleration, and responding to simulated emergencies	-	-	-	-
PC11. use simulator feedback to improve operational skills, focusing on safety and efficiency	-	-	-	-
NOS Total	20	20	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0603
NOS Name	Supervise metro train operations in Non-ATP modes
Sector	Logistics
Sub-Sector	
Occupation	Metro Rail Operations
NSQF Level	4.5
Credits	0.5
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Qualification Pack

DMR/LSC/N0604: Supervise Automated Fare Collection (AFC) system operations and maintenance

Description

This unit outlines the skills and knowledge required to supervise the operations, maintenance, and troubleshooting of Automated Fare Collection (AFC) systems in metro railways, ensuring efficiency and reliability.

Scope

The scope covers the following :

- AFC Gate
- Ticket Office Machine (TOM)
- Ticket Vending Machine (TVM)
- Station Control (SC) Computer
- AFC Media and Developments
- AVM, HHT, RCTM/SCVM
- AFC Practical

Elements and Performance Criteria

AFC Gate

To be competent, the user/individual on the job must be able to:

- PC1.** identify types of AFC gates and their components, including nomenclature and functionalities
- PC2.** access and utilize the maintenance menu for diagnostics and routine checks
- PC3.** execute the container replacement procedure accurately, ensuring secure handling of components
- PC4.** follow operational precautions to avoid system or component damage during maintenance and operation

Ticket Office Machine (TOM)

To be competent, the user/individual on the job must be able to:

- PC5.** identify the types of TOMs and functions of their parts
- PC6.** operate the Man-Machine Interface (MMI) effectively for transaction processing and troubleshooting

Ticket Vending Machine (TVM)

To be competent, the user/individual on the job must be able to:

- PC7.** supervise the operation of TVMs, ensuring smooth transaction processing and resolving common operational issues

Station Control (SC) Computer

To be competent, the user/individual on the job must be able to:

- PC8.** supervise SC computer functions and monitor AFC system performance
- PC9.** execute procedures for changing gate and fare modes using the SC computer

Qualification Pack

PC10. implement emergency commands and verify the effectiveness of the action

PC11. access and analyze AFC reports from the SC computer for performance insights

AFC Media and Developments

To be competent, the user/individual on the job must be able to:

PC12. analyze different AFC media, including QR codes, and identify their operational processes

PC13. interpret error codes and resolve media-related issues effectively

PC14. supervise and integrate the latest technological developments into AFC operations

PC15. assess and manage different gate and fare modes to optimize system usability

AVM, HHT, RCTM/SCVM

To be competent, the user/individual on the job must be able to:

PC16. supervise and ensure proper functioning of Add Value Machines (AVM) and Handheld Terminals (HHT)

PC17. monitor operations of Rechargeable Card Ticket Machines (RCTM) and Smart Card Vending Machines (SCVM)

AFC Practical

To be competent, the user/individual on the job must be able to:

PC18. operate AFC gates, TOMs, TVMs, SC computers, and AFC media

PC19. conduct system inspections and recommend maintenance or corrective actions

PC20. facilitate practical training sessions for team members on AFC systems

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the types of AFC gates, parts nomenclature, maintenance menu functionalities, container replacement procedure, operational precautions

KU2. different types, parts, functions of TOM, and MMI description

KU3. the operational description, troubleshooting steps, and user guidance for TVM

KU4. SC computer supervision functions, gate/fare mode modification, emergency command execution, and report generation

KU5. AFC media types, QR codes, error code interpretation, latest advancements, gate/fare modes

KU6. the functions and troubleshooting of AVM and HHT

KU7. RCTM and SCVM working principles, common issues, and resolution techniques

KU8. the emerging technologies in AFC systems, including AI, contactless payment methods, and media analysis techniques

KU9. the operational safety precautions and incident management procedures

KU10. the interdependencies of AFC components for efficient system operations

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. convey clear and concise information to teams, supervisors, and customers

Qualification Pack

- GS2.** effectively communicate operational protocols and issues to team members and stakeholders
- GS3.** analyze and resolve AFC system issues promptly
- GS4.** document operational processes, troubleshooting steps, and maintenance logs accurately
- GS5.** lead and coordinate a team for AFC system operations and maintenance tasks
- GS6.** ensure precision in operational checks, maintenance, and troubleshooting activities
- GS7.** prioritize and manage time effectively during peak operational hours
- GS8.** ensure AFC systems provide a seamless user experience for commuters
- GS9.** adapt to the adoption of new AFC technologies and updates
- GS10.** evaluate system performance and recommend optimizations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>AFC Gate</i>	3	3	-	2
PC1. identify types of AFC gates and their components, including nomenclature and functionalities	-	-	-	-
PC2. access and utilize the maintenance menu for diagnostics and routine checks	-	-	-	-
PC3. execute the container replacement procedure accurately, ensuring secure handling of components	-	-	-	-
PC4. follow operational precautions to avoid system or component damage during maintenance and operation	-	-	-	-
<i>Ticket Office Machine (TOM)</i>	3	3	-	2
PC5. identify the types of TOMs and functions of their parts	-	-	-	-
PC6. operate the Man-Machine Interface (MMI) effectively for transaction processing and troubleshooting	-	-	-	-
<i>Ticket Vending Machine (TVM)</i>	2	2	-	2
PC7. supervise the operation of TVMs, ensuring smooth transaction processing and resolving common operational issues	-	-	-	-
<i>Station Control (SC) Computer</i>	3	3	-	1
PC8. supervise SC computer functions and monitor AFC system performance	-	-	-	-
PC9. execute procedures for changing gate and fare modes using the SC computer	-	-	-	-
PC10. implement emergency commands and verify the effectiveness of the action	-	-	-	-
PC11. access and analyze AFC reports from the SC computer for performance insights	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>AFC Media and Developments</i>	3	3	-	1
PC12. analyze different AFC media, including QR codes, and identify their operational processes	-	-	-	-
PC13. interpret error codes and resolve media-related issues effectively	-	-	-	-
PC14. supervise and integrate the latest technological developments into AFC operations	-	-	-	-
PC15. assess and manage different gate and fare modes to optimize system usability	-	-	-	-
<i>AVM, HHT, RCTM/SCVM</i>	3	3	-	1
PC16. supervise and ensure proper functioning of Add Value Machines (AVM) and Handheld Terminals (HHT)	-	-	-	-
PC17. monitor operations of Rechargeable Card Ticket Machines (RCTM) and Smart Card Vending Machines (SCVM)	-	-	-	-
<i>AFC Practical</i>	3	3	-	1
PC18. operate AFC gates, TOMs, TVMs, SC computers, and AFC media	-	-	-	-
PC19. conduct system inspections and recommend maintenance or corrective actions	-	-	-	-
PC20. facilitate practical training sessions for team members on AFC systems	-	-	-	-
NOS Total	20	20	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0604
NOS Name	Supervise Automated Fare Collection (AFC) system operations and maintenance
Sector	Logistics
Sub-Sector	
Occupation	Metro Rail Operations
NSQF Level	4.5
Credits	0.6
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DMR/LSC/N0605: Supervise metro station operations

Description

This unit outlines the skills and knowledge required to supervise and manage all aspects of station operations, ensuring safety, efficiency, and passenger satisfaction while maintaining compliance with standard procedures.

Scope

The scope covers the following :

- Station Working Order
- Station Layout
- Traction Sectioning Diagram
- Track Layout
- Essential Equipment & SCR Operations
- Business Rules & Revenue Management
- Public Complaint Handling
- SOP Compliance
- Station Documents & Records
- Defect Recording & Reporting
- Contract Management
- Earthing Rod Placement
- Case Studies & Lessons Learned
- Instructions (Station)
- Manual Point Operation
- Escalator/Travelator Operations
- Lift/Elevator Operations
- ECS/TVS/BMS Operations
- Computer-Based Training (CBT)

Elements and Performance Criteria

Station Working Order

To be competent, the user/individual on the job must be able to:

- PC1.** implement and monitor station working orders to ensure seamless operations
- PC2.** conduct regular checks to confirm compliance with SWO provisions
- PC3.** resolve operational deviations promptly

Station Layout

To be competent, the user/individual on the job must be able to:

- PC4.** interpret station layouts for underground, elevated, and at-grade stations
- PC5.** optimize station area utilization for passenger safety and convenience
- PC6.** compare operational differences between various station types to plan resources effectively

Traction Sectioning Diagram

To be competent, the user/individual on the job must be able to:

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- PC7.** read and interpret traction sectioning diagrams associated with power block operations
- PC8.** coordinate with the traction team to ensure safe execution of power blocks
- PC9.** maintain updated traction diagrams for reference and training

Track Layout

To be competent, the user/individual on the job must be able to:

- PC10.** monitor station track layouts, including points, signals, and crossovers, for operational readiness
- PC11.** identify and report track-related anomalies or safety concerns
- PC12.** coordinate with the track maintenance team for timely issue resolution

Essential Equipment & SCR Operations

To be competent, the user/individual on the job must be able to:

- PC13.** operate and troubleshoot essential SCR equipment, ensuring all systems are functional
- PC14.** conduct periodic maintenance checks on SCR workstations and equipment
- PC15.** train station staff on the use of critical SCR equipment

Business Rules & Revenue Management

To be competent, the user/individual on the job must be able to:

- PC16.** enforce adherence to business rules and revenue circulars
- PC17.** oversee the collection, reconciliation, and reporting of station revenue
- PC18.** supervise lost-and-found item management as per standard protocols
- PC19.** generate and submit accurate revenue reports, including periodical returns and balance sheets

Public Complaint Handling

To be competent, the user/individual on the job must be able to:

- PC20.** address public complaints efficiently and document resolutions
- PC21.** implement corrective actions to minimize recurring issues
- PC22.** maintain a database of complaint trends for continuous improvement

SOP Compliance

To be competent, the user/individual on the job must be able to:

- PC23.** ensure adherence to SOPs in all station operations
- PC24.** conduct regular reviews and updates of SOPs based on operational feedback

Station Documents & Records

To be competent, the user/individual on the job must be able to:

- PC25.** maintain station records, including safety forms, revenue documents, and defect logs
- PC26.** ensure timely and accurate online reporting of station documents and registers
- PC27.** conduct audits to confirm the completeness and accuracy of records

Defect Recording & Reporting

To be competent, the user/individual on the job must be able to:

- PC28.** record operational defects promptly and escalate them to relevant departments
- PC29.** track reported defects to ensure timely resolution
- PC30.** compile defect data for performance analysis and reporting

Contract Management

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To be competent, the user/individual on the job must be able to:

- PC31.** supervise contracts related to ticketing, customer facilitation agents, parking, and housekeeping
- PC32.** monitor contractor compliance with SLAs and escalate breaches appropriately
- PC33.** review and update contract-related records periodically

Earthing Rod Placement

To be competent, the user/individual on the job must be able to:

- PC34.** place earthing rods safely and efficiently during maintenance or emergencies
- PC35.** verify earthing rod placement before commencing related activities

Case Studies & Lessons Learned

To be competent, the user/individual on the job must be able to:

- PC36.** analyze case studies related to station failures and implement lessons learned
- PC37.** conduct team discussions on failure analysis to enhance station operations

Instructions (Station)

To be competent, the user/individual on the job must be able to:

- PC38.** communicate station-specific instructions clearly to the team
- PC39.** monitor adherence to issued instructions and provide guidance as needed

Manual Point Operation

To be competent, the user/individual on the job must be able to:

- PC40.** operate manual points during emergencies following standard procedures
- PC41.** ensure proper communication with OCC during manual point operations

Escalator/Travelator Operations

To be competent, the user/individual on the job must be able to:

- PC42.** operate escalators and travelators, including emergency start/stop mechanisms
- PC43.** conduct regular safety checks on escalators/travelators
- PC44.** train station staff on proper escalator and travelator usage and safety

Lift/Elevator Operations

To be competent, the user/individual on the job must be able to:

- PC45.** operate lifts, including emergency rescue procedures, adhering to safety protocols
- PC46.** conduct periodic inspections to ensure lift functionality and safety
- PC47.** train staff on lift operations and passenger rescue techniques

ECS/TVS/BMS Operations

To be competent, the user/individual on the job must be able to:

- PC48.** operate Environmental Control Systems (ECS), Tunnel Ventilation Systems (TVS), and Building Management Systems (BMS) effectively
- PC49.** utilize system features to maintain passenger safety and station comfort
- PC50.** report malfunctions in ECS/TVS/BMS to the maintenance team promptly

Computer-Based Training (CBT)

To be competent, the user/individual on the job must be able to:

- PC51.** conduct computer-based training sessions for station management processes
- PC52.** evaluate staff proficiency through CBT modules and provide feedback

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the station working order guidelines and compliance procedures
- KU2.** the differences in station layouts for underground, elevated, and at-grade stations
- KU3.** the components of traction sectioning diagrams and their operational significance
- KU4.** track layout elements such as points, signals, and crossovers
- KU5.** the business rules, revenue circulars, and cash handling processes
- KU6.** the functions and troubleshooting of SCR equipment
- KU7.** the protocols for managing public complaints and lost-and-found items
- KU8.** the escalator, travelator, and lift safety features and operation
- KU9.** ECS, TVS, and BMS functionalities and maintenance requirements
- KU10.** manual point operations and earthing rod placement
- KU11.** case study analysis for station failure scenarios
- KU12.** the use of online tools for station records and reporting

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** convey clear and concise information to teams, supervisors, and customers
- GS2.** effectively communicate instructions, updates, and feedback to staff and passengers
- GS3.** address operational issues and identify preventive measures
- GS4.** make informed decisions during emergencies or disruptions
- GS5.** lead station teams to achieve operational excellence
- GS6.** operate and maintain station control systems and equipment
- GS7.** resolve disputes among passengers or staff diplomatically
- GS8.** prioritize tasks to ensure timely completion
- GS9.** maintain accurate records and monitor operational adherence
- GS10.** adjust to dynamic situations and implement innovative solutions
- GS11.** enhance passenger satisfaction through proactive assistance and complaint resolution

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Station Working Order</i>	1	1	-	1
PC1. implement and monitor station working orders to ensure seamless operations	-	-	-	-
PC2. conduct regular checks to confirm compliance with SWO provisions	-	-	-	-
PC3. resolve operational deviations promptly	-	-	-	-
<i>Station Layout</i>	1	1	-	0.5
PC4. interpret station layouts for underground, elevated, and at-grade stations	-	-	-	-
PC5. optimize station area utilization for passenger safety and convenience	-	-	-	-
PC6. compare operational differences between various station types to plan resources effectively	-	-	-	-
<i>Traction Sectioning Diagram</i>	1	1	-	0.5
PC7. read and interpret traction sectioning diagrams associated with power block operations	-	-	-	-
PC8. coordinate with the traction team to ensure safe execution of power blocks	-	-	-	-
PC9. maintain updated traction diagrams for reference and training	-	-	-	-
<i>Track Layout</i>	1	1	-	0.5
PC10. monitor station track layouts, including points, signals, and crossovers, for operational readiness	-	-	-	-
PC11. identify and report track-related anomalies or safety concerns	-	-	-	-
PC12. coordinate with the track maintenance team for timely issue resolution	-	-	-	-
<i>Essential Equipment & SCR Operations</i>	1	1	-	0.5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. operate and troubleshoot essential SCR equipment, ensuring all systems are functional	-	-	-	-
PC14. conduct periodic maintenance checks on SCR workstations and equipment	-	-	-	-
PC15. train station staff on the use of critical SCR equipment	-	-	-	-
<i>Business Rules & Revenue Management</i>	2	2	-	0.5
PC16. enforce adherence to business rules and revenue circulars	-	-	-	-
PC17. oversee the collection, reconciliation, and reporting of station revenue	-	-	-	-
PC18. supervise lost-and-found item management as per standard protocols	-	-	-	-
PC19. generate and submit accurate revenue reports, including periodical returns and balance sheets	-	-	-	-
<i>Public Complaint Handling</i>	1	1	-	0.5
PC20. address public complaints efficiently and document resolutions	-	-	-	-
PC21. implement corrective actions to minimize recurring issues	-	-	-	-
PC22. maintain a database of complaint trends for continuous improvement	-	-	-	-
<i>SOP Compliance</i>	1	1	-	0.5
PC23. ensure adherence to SOPs in all station operations	-	-	-	-
PC24. conduct regular reviews and updates of SOPs based on operational feedback	-	-	-	-
<i>Station Documents & Records</i>	1	1	-	0.5
PC25. maintain station records, including safety forms, revenue documents, and defect logs	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC26. ensure timely and accurate online reporting of station documents and registers	-	-	-	-
PC27. conduct audits to confirm the completeness and accuracy of records	-	-	-	-
<i>Defect Recording & Reporting</i>	1	1	-	0.5
PC28. record operational defects promptly and escalate them to relevant departments	-	-	-	-
PC29. track reported defects to ensure timely resolution	-	-	-	-
PC30. compile defect data for performance analysis and reporting	-	-	-	-
<i>Contract Management</i>	1	1	-	0.5
PC31. supervise contracts related to ticketing, customer facilitation agents, parking, and housekeeping	-	-	-	-
PC32. monitor contractor compliance with SLAs and escalate breaches appropriately	-	-	-	-
PC33. review and update contract-related records periodically	-	-	-	-
<i>Earthing Rod Placement</i>	1	1	-	0.5
PC34. place earthing rods safely and efficiently during maintenance or emergencies	-	-	-	-
PC35. verify earthing rod placement before commencing related activities	-	-	-	-
<i>Case Studies & Lessons Learned</i>	1	1	-	0.5
PC36. analyze case studies related to station failures and implement lessons learned	-	-	-	-
PC37. conduct team discussions on failure analysis to enhance station operations	-	-	-	-
<i>Instructions (Station)</i>	1	1	-	0.5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC38. communicate station-specific instructions clearly to the team	-	-	-	-
PC39. monitor adherence to issued instructions and provide guidance as needed	-	-	-	-
<i>Manual Point Operation</i>	1	1	-	0.5
PC40. operate manual points during emergencies following standard procedures	-	-	-	-
PC41. ensure proper communication with OCC during manual point operations	-	-	-	-
<i>Escalator/Travelator Operations</i>	1	1	-	0.5
PC42. operate escalators and travelators, including emergency start/stop mechanisms	-	-	-	-
PC43. conduct regular safety checks on escalators/travelators	-	-	-	-
PC44. train station staff on proper escalator and travelator usage and safety	-	-	-	-
<i>Lift/Elevator Operations</i>	1	1	-	0.5
PC45. operate lifts, including emergency rescue procedures, adhering to safety protocols	-	-	-	-
PC46. conduct periodic inspections to ensure lift functionality and safety	-	-	-	-
PC47. train staff on lift operations and passenger rescue techniques	-	-	-	-
<i>ECS/TVS/BMS Operations</i>	1	1	-	0.5
PC48. operate Environmental Control Systems (ECS), Tunnel Ventilation Systems (TVS), and Building Management Systems (BMS) effectively	-	-	-	-
PC49. utilize system features to maintain passenger safety and station comfort	-	-	-	-
PC50. report malfunctions in ECS/TVS/BMS to the maintenance team promptly	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Computer-Based Training (CBT)</i>	1	1	-	0.5
PC51. conduct computer-based training sessions for station management processes	-	-	-	-
PC52. evaluate staff proficiency through CBT modules and provide feedback	-	-	-	-
NOS Total	20	20	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0605
NOS Name	Supervise metro station operations
Sector	Logistics
Sub-Sector	
Occupation	Metro Rail Operations
NSQF Level	4.5
Credits	3.4
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DMR/LSC/N0606: Supervise and troubleshoot rolling stock

Description

This unit outlines the competencies required to oversee and ensure efficient operation, troubleshooting, and maintenance of rolling stock systems, ensuring the reliability of metro rail services

Scope

The scope covers the following :

- Rolling Stock Systems
- Train Subsystems
- Operating and Troubleshooting Procedures
- Rolling Stock Practical Exposure
- Simulator Training

Elements and Performance Criteria

Rolling Stock Systems

To be competent, the user/individual on the job must be able to:

- PC1.** inspect rolling stock cab equipment and verify their functionality
- PC2.** check and report the condition of internal and external rolling stock equipment, including the bogie, primary suspension, and couplers
- PC3.** ensure proper functioning of safety-critical equipment during routine inspections

Train Subsystems

To be competent, the user/individual on the job must be able to:

- PC4.** monitor the performance of the Train Control Management System (TCMS)
- PC5.** supervise current collection systems, propulsion systems, and auxiliary systems, including battery systems and train lighting
- PC6.** coordinate with the maintenance team for troubleshooting HVAC systems, door systems, PA-PIS, CCTV, and passenger alarm devices
- PC7.** verify operational readiness of the air supply and brake systems, ensuring correct functioning of motoring, braking, brake loops, and emergency brake loops

Operating and Troubleshooting Procedures

To be competent, the user/individual on the job must be able to:

- PC8.** follow established guidelines for operating rolling stock and troubleshooting common issues
- PC9.** respond to system malfunctions by implementing standard emergency and corrective actions
- PC10.** document all troubleshooting activities and escalate unresolved issues to higher technical authorities

Rolling Stock Practical Exposure

To be competent, the user/individual on the job must be able to:

- PC11.** train on rolling stock equipment to gain hands-on experience

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PC12. use diagnostic tools following safety procedures during maintenance tasks

PC13. supervise the execution of practical tasks to ensure adherence to safety and operational standards

Simulator Training

To be competent, the user/individual on the job must be able to:

PC14. familiarize with the cab, including TCMS and control systems, using simulators

PC15. simulate various train driving scenarios to enhance problem-solving skills and emergency response readiness

PC16. evaluate the performance of team members in handling simulated scenarios and provide constructive feedback

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the types and functionalities of train control systems, including ATP and non-ATP modes

KU2. the components of rolling stock, including cab equipment, bogie, suspension systems, and couplers

KU3. the functions of internal and external rolling stock equipment

KU4. the principles of TCMS, current collection, propulsion systems, auxiliary systems, and battery systems

KU5. the operation of HVAC, door systems, PA-PIS, CCTV, passenger alarm devices, air supply systems, and brake systems

KU6. the safety protocols for motoring, braking, and brake loops

KU7. the standard operating procedures for rolling stock

KU8. the common troubleshooting methods and fault diagnosis techniques

KU9. the safety guidelines for emergency and corrective actions

KU10. the use of appropriate diagnostic tools and equipment for rolling stock maintenance

KU11. the techniques for inspecting and maintaining critical rolling stock systems

KU12. the simulator features, including TCMS and cab controls

KU13. the emergency response protocols simulated for practical understanding

KU14. the methods for evaluating and improving driving techniques

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. analyze system performance to identify faults in rolling stock systems

GS2. resolve technical issues promptly to minimize service disruptions

GS3. convey technical information clearly to the team and other stakeholders

GS4. lead maintenance and troubleshooting teams effectively

GS5. prioritize tasks during emergency and routine operations

GS6. adjust to new technologies and operating procedures for rolling stock systems

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- GS7.** apply and enforce safety protocols consistently
- GS8.** record maintenance activities accurately for compliance and reporting
- GS9.** conduct practical and simulator-based training sessions
- GS10.** monitor equipment conditions and detect potential issues early

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Rolling Stock Systems</i>	4	4	-	2
PC1. inspect rolling stock cab equipment and verify their functionality	-	-	-	-
PC2. check and report the condition of internal and external rolling stock equipment, including the bogie, primary suspension, and couplers	-	-	-	-
PC3. ensure proper functioning of safety-critical equipment during routine inspections	-	-	-	-
<i>Train Subsystems</i>	4	4	-	2
PC4. monitor the performance of the Train Control Management System (TCMS)	-	-	-	-
PC5. supervise current collection systems, propulsion systems, and auxiliary systems, including battery systems and train lighting	-	-	-	-
PC6. coordinate with the maintenance team for troubleshooting HVAC systems, door systems, PA-PIS, CCTV, and passenger alarm devices	-	-	-	-
PC7. verify operational readiness of the air supply and brake systems, ensuring correct functioning of motoring, braking, brake loops, and emergency brake loops	-	-	-	-
<i>Operating and Troubleshooting Procedures</i>	4	4	-	2
PC8. follow established guidelines for operating rolling stock and troubleshooting common issues	-	-	-	-
PC9. respond to system malfunctions by implementing standard emergency and corrective actions	-	-	-	-
PC10. document all troubleshooting activities and escalate unresolved issues to higher technical authorities	-	-	-	-
<i>Rolling Stock Practical Exposure</i>	4	4	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. train on rolling stock equipment to gain hands-on experience	-	-	-	-
PC12. use diagnostic tools following safety procedures during maintenance tasks	-	-	-	-
PC13. supervise the execution of practical tasks to ensure adherence to safety and operational standards	-	-	-	-
<i>Simulator Training</i>	4	4	-	2
PC14. familiarize with the cab, including TCMS and control systems, using simulators	-	-	-	-
PC15. simulate various train driving scenarios to enhance problem-solving skills and emergency response readiness	-	-	-	-
PC16. evaluate the performance of team members in handling simulated scenarios and provide constructive feedback	-	-	-	-
NOS Total	20	20	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0606
NOS Name	Supervise and troubleshoot rolling stock
Sector	Logistics
Sub-Sector	
Occupation	Metro Rail Operations
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DMR/LSC/N0607: Supervise SOP train operations and practical tasks

Description

This unit outlines the competencies required to supervise and ensure efficient train operations, including handling failures, troubleshooting, rescue operations, and adherence to SOPs, while utilizing technical tools like simulators and CBT modules

Scope

The scope covers the following :

- Failure Analysis and Technical Reporting
- Simulator-Based Communication and Troubleshooting
- Standard Operating Procedure (SOP) for Train Operations

Elements and Performance Criteria

Failure Analysis and Technical Reporting

To be competent, the user/individual on the job must be able to:

- PC1.** identify causes of train operation failures using diagnostic tools and data logs
- PC2.** categorize failures into mechanical, electrical, or operational domains
- PC3.** draft comprehensive Failure Analysis Reports (FAR) and technical instructions
- PC4.** submit recommendations for corrective and preventive actions
- PC5.** maintain updated records of FARs and technical instructions for reference

Simulator-Based Communication and Troubleshooting

To be competent, the user/individual on the job must be able to:

- PC6.** operate simulators to replicate train scenarios for troubleshooting and rescue drills
- PC7.** conduct communication drills with signaling and control teams
- PC8.** identify and address random faults simulated in the Control-Based Trainer (CBT)
- PC9.** practice quick decision-making under simulated fault conditions
- PC10.** execute mock rescue operations to test preparedness and adherence to protocols
- PC11.** coordinate with relevant teams during rescue simulation drills

Standard Operating Procedure (SOP) for Train Operations

To be competent, the user/individual on the job must be able to:

- PC12.** supervise the implementation of SOPs during live and simulated operations
- PC13.** monitor operational parameters to ensure compliance with safety and efficiency standards
- PC14.** conduct practical sessions for team members on SOPs and train operations
- PC15.** provide feedback and evaluate the competency of team members in handling operational tasks
- PC16.** conduct periodic SOP compliance audits
- PC17.** recommend updates to SOPs based on operational feedback and incident analysis

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the types of metro trains, their components, and basic operational principles
- KU2.** the train control systems, fault diagnostic software, and simulators
- KU3.** the guidelines for drafting technical instructions and failure analysis reports
- KU4.** the protocols for communication between train operators, signaling teams, and rescue staff
- KU5.** the DMRC SOPs, safety standards, and emergency protocols
- KU6.** the rescue operations, hazard management, and passenger safety measures
- KU7.** the approaches to mechanical and electrical fault rectification
- KU8.** the techniques for conducting simulator-based training and evaluating performance

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** analyze failure patterns to determine root causes and solutions
- GS2.** address random faults and emergency scenarios effectively
- GS3.** monitor operational parameters and identify deviations
- GS4.** work closely with other teams during rescue and troubleshooting operations
- GS5.** make quick, informed decisions in high-pressure situations
- GS6.** operate simulators and diagnostic tools confidently
- GS7.** guide team members during practical drills and SOP training
- GS8.** maintain accurate records of reports and instructions
- GS9.** handle evolving operational challenges and update practices as needed

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Failure Analysis and Technical Reporting</i>	7	7	-	3
PC1. identify causes of train operation failures using diagnostic tools and data logs	-	-	-	-
PC2. categorize failures into mechanical, electrical, or operational domains	-	-	-	-
PC3. draft comprehensive Failure Analysis Reports (FAR) and technical instructions	-	-	-	-
PC4. submit recommendations for corrective and preventive actions	-	-	-	-
PC5. maintain updated records of FARs and technical instructions for reference	-	-	-	-
<i>Simulator-Based Communication and Troubleshooting</i>	7	7	-	3
PC6. operate simulators to replicate train scenarios for troubleshooting and rescue drills	-	-	-	-
PC7. conduct communication drills with signaling and control teams	-	-	-	-
PC8. identify and address random faults simulated in the Control-Based Trainer (CBT)	-	-	-	-
PC9. practice quick decision-making under simulated fault conditions	-	-	-	-
PC10. execute mock rescue operations to test preparedness and adherence to protocols	-	-	-	-
PC11. coordinate with relevant teams during rescue simulation drills	-	-	-	-
<i>Standard Operating Procedure (SOP) for Train Operations</i>	6	6	-	4
PC12. supervise the implementation of SOPs during live and simulated operations	-	-	-	-
PC13. monitor operational parameters to ensure compliance with safety and efficiency standards	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. conduct practical sessions for team members on SOPs and train operations	-	-	-	-
PC15. provide feedback and evaluate the competency of team members in handling operational tasks	-	-	-	-
PC16. conduct periodic SOP compliance audits	-	-	-	-
PC17. recommend updates to SOPs based on operational feedback and incident analysis	-	-	-	-
NOS Total	20	20	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0607
NOS Name	Supervise SOP train operations and practical tasks
Sector	Logistics
Sub-Sector	
Occupation	Metro Rail Operations
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DMR/LSC/N0608: Operate metro train on test track, mainline (non-revenue hours), and simulator

Description

This unit outlines the competencies required to supervise and perform cab operations, including driving practice on test tracks, mainline during non-revenue hours, and simulator-based driving to ensure operational efficiency and adherence to safety protocols

Scope

The scope covers the following :

- Test Track / Mainline (Non-Revenue Hours) Driving
- Simulator Driving
- Cab Exposure (On Mainline)

Elements and Performance Criteria

Test Track / Mainline (Non-Revenue Hours) Driving

To be competent, the user/individual on the job must be able to:

- PC1.** conduct driving operations on test tracks and mainline during non-revenue hours
- PC2.** inspect the train and track conditions before initiating the drive
- PC3.** monitor speed, signals, and communication systems during operation
- PC4.** coordinate with the Operations Control Centre (OCC) for real-time instructions and incident reporting
- PC5.** ensure compliance with safety protocols during non-revenue operations
- PC6.** document driving logs, safety checks, and incidents encountered

Simulator Driving

To be competent, the user/individual on the job must be able to:

- PC7.** practice and refine driving skills using the simulator
- PC8.** operate the simulator to simulate real-life driving conditions, including emergencies
- PC9.** analyze and interpret feedback from simulator sessions to identify areas of improvement
- PC10.** train subordinate staff on simulator operations and assess their driving competencies
- PC11.** maintain the simulator logbook and coordinate with the training department for updates

Cab Exposure (On Mainline)

To be competent, the user/individual on the job must be able to:

- PC12.** conduct supervised cab exposure sessions for new drivers or trainees on the mainline
- PC13.** evaluate driver competencies during exposure and provide corrective instructions
- PC14.** familiarize with signaling systems, track layouts, and operational scenarios on the mainline
- PC15.** ensure adherence to operational guidelines and safety regulations during cab exposure
- PC16.** report performance and observations to the training department or higher management

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the types of metro trains, their components, and basic operational principles
- KU2.** the working principles of metro trains and key systems (propulsion, braking, signaling)
- KU3.** the safety protocols for test track and mainline operations
- KU4.** the functionality of simulator equipment and training modules
- KU5.** the emergency response procedures, including track-side incidents
- KU6.** DMRC operational guidelines and SOPs
- KU7.** traffic regulations and signaling systems for metro railways
- KU8.** incident and safety reporting formats
- KU9.** the procedures for mainline driving during non-revenue hours
- KU10.** the communication protocols with OCC and other operational teams
- KU11.** the maintenance reporting for cab systems and simulator equipment
- KU12.** the methods for evaluating driver performance and identifying training needs
- KU13.** the conflict resolution techniques and team management principles
- KU14.** the documentation standards for operational logs and training records

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively with OCC, subordinates, and trainees to ensure smooth operations
- GS2.** lead and guide teams during cab exposure and simulator sessions
- GS3.** make prompt and informed decisions in operational scenarios
- GS4.** assess and resolve operational and training challenges
- GS5.** respond effectively to changing conditions during non-revenue driving and simulator operations
- GS6.** monitor train systems, signals, and driver actions to ensure precision
- GS7.** address technical and operational issues proactively
- GS8.** maintain accurate logs for operations, safety checks, and simulator sessions
- GS9.** instill a culture of safety during all operations and training sessions
- GS10.** operate simulator systems and train control panels efficiently

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Test Track / Mainline (Non-Revenue Hours) Driving</i>	7	7	-	3
PC1. conduct driving operations on test tracks and mainline during non-revenue hours	-	-	-	-
PC2. inspect the train and track conditions before initiating the drive	-	-	-	-
PC3. monitor speed, signals, and communication systems during operation	-	-	-	-
PC4. coordinate with the Operations Control Centre (OCC) for real-time instructions and incident reporting	-	-	-	-
PC5. ensure compliance with safety protocols during non-revenue operations	-	-	-	-
PC6. document driving logs, safety checks, and incidents encountered	-	-	-	-
<i>Simulator Driving</i>	7	7	-	3
PC7. practice and refine driving skills using the simulator	-	-	-	-
PC8. operate the simulator to simulate real-life driving conditions, including emergencies	-	-	-	-
PC9. analyze and interpret feedback from simulator sessions to identify areas of improvement	-	-	-	-
PC10. train subordinate staff on simulator operations and assess their driving competencies	-	-	-	-
PC11. maintain the simulator logbook and coordinate with the training department for updates	-	-	-	-
<i>Cab Exposure (On Mainline)</i>	6	6	-	4
PC12. conduct supervised cab exposure sessions for new drivers or trainees on the mainline	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. evaluate driver competencies during exposure and provide corrective instructions	-	-	-	-
PC14. familiarize with signaling systems, track layouts, and operational scenarios on the mainline	-	-	-	-
PC15. ensure adherence to operational guidelines and safety regulations during cab exposure	-	-	-	-
PC16. report performance and observations to the training department or higher management	-	-	-	-
NOS Total	20	20	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0608
NOS Name	Operate metro train on test track, mainline (non-revenue hours), and simulator
Sector	Logistics
Sub-Sector	
Occupation	Metro Rail Operations
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on a knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training centre (as per the assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practicals for every student at each examination/training centre based on this criterion.
6. To pass the Qualification Pack, every trainee should score a minimum of 60% of aggregate marks to successfully clear the assessment.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 60

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
DMR/LSC/N9901.Implement metro operations training and safety procedures	20	20	0	10	50	10
DMR/LSC/N0601.Supervise the operations of signaling systems and platform screen doors	20	20	0	10	50	20
DMR/LSC/N0602.Supervise Telecom and Communication Systems in Metro Railways	20	20	0	10	50	20
DMR/LSC/N0603.Supervise metro train operations in Non-ATP modes	20	20	0	10	50	20
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	100	110	0	40	250	80

Elective: 1 Station Controller

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
DMR/LSC/N0604.Supervise Automated Fare Collection (AFC) system operations and maintenance	20	20	0	10	50	10

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
DMR/LSC/N0605.Supervise metro station operations	20	20	0	10	50	10
Total	40	40	-	20	100	20

Elective: 2 Train Operator

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
DMR/LSC/N0606.Supervise and troubleshoot rolling stock	20	20	0	10	50	10
DMR/LSC/N0607.Supervise SOP train operations and practical tasks	20	20	0	10	50	5
DMR/LSC/N0608.Operate metro train on test track, mainline (non-revenue hours), and simulator	20	20	0	10	50	5
Total	60	60	-	30	150	20

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NCVET	National Council for Vocational Education and Training
QP	Qualification Pack
MC	Model Curriculum
NSQF	National Skills Qualification Framework
NSQC	National Skills Qualification Committee
NOS	National Occupational Standards
NCO	National Classification of Occupations
ES	Employability Skills
MR	Metro Railway
DMRC	Delhi Metro Rail Corporation
DMRA	Delhi Metro Rail Academy
AFC	Automatic Fare Collection
ATS	Automatic Train Supervision
ATP	Automatic Train Protection
S&T	Signalling and Telecommunications
CBTC	Communications-Based Train Control
RRI	Relay-Interlocking
SSI	Solid State Interlocking
CBI	Computer-Based Interlocking
PSD	Platform Screen Doors
E&M	Electrical and Mechanica

Qualification Pack

OHE	Overhead Electrification
SCADA	Supervisory Control and Data Acquisition
TVS	Train Vision System
OTDR	Optical Time Domain Reflectometer
SDH	Synchronous Digital Hierarchy
PDH	Plesiochronous Digital Hierarchy
MPLS	Multiprotocol Label Switching
PIDS	Passenger Information Display System
PAS	Public Announcement System
IPBX	Internet Protocol Private Branch Exchange
OCC	Operations Control Center
NCMC	National Common Mobility Card
TVMs	Ticket Vending Machines
TOM	Ticket Office Machine
EFO	Electronic Fare Organizer
TR/AVR	Ticket Reader/Add Value Machine
SC/CC	Station Computer/ Central Computer
CCHS	Central Clearing House System

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.